WANG, Yankai （ワン・イエンカイ）

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Japanese (Professional working, JLPT N1 121/180) · English (Professional, TOEFL 97/120) · Chinese (Native)

Education

The University of Tokyo 東京大学 — Oct. 2025 – Sep. 2027 (Expected)

M.S. Candidate, Information & Communication Engineering — Tokyo, Japan

- Suzumura Laboratory; pursuing research on **Vision-Language-Action (VLA)** models and human-robot collaboration benchmarks for scientific laboratory environments.

Shanghai University 上海大学 — Sep. 2021 – Jul. 2025

B.E., Electronic Information Engineering — Shanghai, China

Projects

KotobaLab — Apr. 2026 – Present

Personal Project: SwiftUI, GRDB 7, SQLite, SwiftData, Swift Testing, iOS 18+ — Tokyo, Japan

Local-first Japanese dictionary app built with SwiftUI, MVVM, and a Clean Architecture–inspired layered design. Currently in local development.

- Built a local-first dictionary app with a **293K-entry, 52 MB SQLite** database generated by a Python pipeline, bundled as an app resource and queried through GRDB `DatabaseQueue`.
- Optimized prefix search with `PRAGMA case_sensitive_like=ON`, improving benchmarked queries from ∼16 ms table scans to **∼0.03 ms** multi-index lookups.
- Designed a four-layer architecture (`AppDependencies` → `Scene` → `Store` → `View`) following **MVVM** and **Clean Architecture** principles; applied the **Repository pattern** with protocol-based abstractions and dependency injection to decouple Domain logic from UI and storage frameworks.
- Combined GRDB-managed SQLite for read-heavy reference data with SwiftData for user state; wrote 13 unit tests with Swift Testing covering domain use cases and repository behavior.
- Roadmap: SQLite FTS5 full-text search, actor-isolated repositories under Swift Concurrency, TestFlight beta.
- Github: shiinayane/KotobaLab.

LabMate — Dec. 2025 – Present

Suzumura Lab: Python, Isaac Sim, VLA Models — Tokyo, Japan

Simulation-first benchmark for human-robot collaboration in scientific laboratory environments.

- Extending the **LabUtopia** simulator into a benchmark for natural-language human-robot collaboration in scientific laboratories, focusing on clarification, safety-aware refusal, grounding, and structured action logging.
- Designing evaluation framework to compare planner architectures (**LLM**, **VLA**, and hybrid systems) on multi-step laboratory protocols; sim-first phase planned for 6+ months before hardware integration.
- Coordinating with 3 procured robotic platforms for future hardware integration; prioritizing rigorous evaluation methodology over demonstration-driven research.

Experience

GlucoPI — Feb. 2025 – Jun. 2025

B.E. Graduation Project, **Advisor**: *Qi Zhang*, PhD, Professor — Shanghai, China

WeChat Mini Program for diabetes management, doctor-patient communication, and glucose prediction.

- Built an end-to-end diabetes self-tracking app with WeChat Mini Program frontend, FastAPI backend, MySQL/MongoDB storage, and WebSocket doctor-patient chat.
- Implemented login and permissions, glucose / diet / insulin records, doctor-patient pairing, real-time messaging, and trend charts.
- Added short-horizon glucose prediction in PyTorch by reproducing the GluPred pipeline and running experiments on **OhioT1DM**.

- Integrated a lightweight LLM Q&A feature for non-diagnostic health guidance.
- Github: shiinayane/glucopi.

Arrived or Not

Dec. 2023 – Dec. 2024

Role: Co-Leader, **Advisor**: *Qi Zhang*, PhD, Professor

Shanghai, China

Machine-vision teaching platform for classroom attendance and course management.

- Implemented real-time face-recognition attendance with RetinaFace detection and **ArcFace** embedding-based identity matching.
- Built teacher / student mobile workflows for class setup, attendance records, and basic engagement statistics.
- Developed a Flutter client and FastAPI service with a clear frontend-backend split.
- Github: shiinayane/Arrived-or-Not-Frontend and shiinayane/Arrived-or-Not-Backend.

Interests

Product-First Engineering: Building products that solve real problems; treating iOS, backend, and full-stack as means rather than ends.
iOS Native Architecture: Local-first design with clean Domain / UI separation; currently focused on shipping native iOS apps as the most direct path to product impact.
Research-to-Product: Bridging VLA / LLM research from graduate work to consumer mobile experiences.

Skills

iOS / Apple Platforms: Swift, SwiftUI, GRDB, SQLite, SwiftData, Observation, Swift Concurrency, Swift Testing
Architecture & Design: MVVM, Clean Architecture, Repository Pattern, Dependency Injection, Protocol-Oriented Programming
Backend / Web: Python (FastAPI), JavaScript
ML / Research: PyTorch, VLA Models, Isaac Sim
Tools / Platforms: Git, Docker, Xcode, VS Code, macOS, Linux